PRAVE / RMC Field Decoder Tester

Version 1.0 Gold - Installation, Operation, and Verification

Matching bench generator for the PRAVE / ME / Time Keeper field decoder

1. Purpose

This tester generates one mixed serial stream that behaves like the expected Raveon field input. **$GPRMC** is sent once per second for the local ME position and UTC date/time. **$PRAVE** is sent asynchronously at randomized intervals for six remote units, each carrying a different RSSI test state.

- Exercises the decoder with interleaved RMC and PRAVE traffic on one COM port.
- Creates six PRAVE units so all six fire-truck RSSI icon files can be visible at the same time.
- Uses valid NMEA checksums on every generated sentence.
- Uses the official proven PRAVE field positions: location at indexes 3 and 4, district at 8, RSSI at 12, and individual ID at 13.
- Uses current UTC date and time in every RMC sentence so the future Windows time-keeper path receives realistic data.

2. Exact Files in This Tester ZIP

Filename	Function
PRAVE_RMC_Field_Decoder_Tester_v1_0_Gold.py	Main mixed-stream serial generator.
Run_PRAVE_RMC_Field_Decoder_Tester_v1_0_Gold.bat	One-click Windows launcher.
PRAVE_RMC_Field_Decoder_Tester_v1_0_Gold_Instructions.pdf	Installation, operation, sentence map, and bench checklist.

3. Default Link Layout

Link	Default	Purpose
Tester output	COM3 @ 19200	Mixed RMC and PRAVE stream.
Decoder input	User-selected COM port	Receives the tester through a null-modem, paired FTDI adapters, or a virtual COM pair.
RMC cadence	1.0 second	Local ME position plus UTC date/time.
PRAVE cadence	Random 1.5-3.5 seconds	One unit per event; all six units are guaranteed once per shuffled cycle.

Change only these values near the top of the Python file when needed:

```
OUTPUT_PORT = "COM3"
OUTPUT_BAUD = 19200
PRAVE_MIN_INTERVAL_SECONDS = 1.5
PRAVE_MAX_INTERVAL_SECONDS = 3.5
```

4. Generated RSSI Fleet

The six units form a compact grid around ME. Each unit holds one fixed RSSI state so the decoder can display all icon variants simultaneously.

Display ID	RSSI field	Expected icon	Latitude	Longitude
7-101	blank	unknown	30.422107	-81.929999
7-102	-115 dBm	1 bar	30.422107	-81.926499
7-103	-105 dBm	2 bars	30.422107	-81.922999
7-104	-95 dBm	3 bars	30.416107	-81.929999
7-105	-85 dBm	4 bars	30.416107	-81.926499
7-106	-75 dBm	5 bars	30.416107	-81.922999

Initial Version 1 test bands: 1 bar = -120 through -111 dBm; 2 bars = -110 through -101; 3 bars = -100 through -91; 4 bars = -90 through -81; 5 bars = -80 or stronger. These are configurable bench bands and can be recalibrated after Raveon field testing.

5. Official PRAVE Test Structure

The tester follows the current proven PRAVE structure and includes both optional trailing fields before the checksum.

Split index	Field	Tester value/use
0	$PRAVE	Header
1	Time slot	Four digits
2	Send To	1234
3	Latitude	Signed NMEA-style degrees/minutes
4	Longitude	Signed NMEA-style degrees/minutes
5	UTC	HHMMSS
6	GPS status	2
7	Satellites	12
8	District ID	First ID character
9	ATAI	0 / unused
10	Voltage	12.80
11	I/O	0
12	RSSI	Blank or signed dBm
13	Individual ID	Three digits
14	State ID	0
15-16	Optional	Blank

6. GPRMC Behavior

- One valid $GPRMC sentence is sent every second.
- Status is A (valid).
- The local ME coordinate is 30.419107, -81.926499 by default.
- Speed and course are zero for a stationary bench test.
- UTC time and DDMMYY date come from the tester computer at transmission time.
- The decoder should use RMC for ME plotting, GPS freshness, and Windows UTC synchronization.

7. Normal Bench Setup

1. Extract the ZIP into one dedicated folder. Do not run it from inside the ZIP.
2. Install Python and pyserial on the tester computer.
3. Connect the tester output COM port to the field decoder input through a true null-modem path, paired FTDI adapters, or a virtual COM-port pair.
4. Set OUTPUT_PORT and OUTPUT_BAUD at the top of the tester Python file.
5. Set the decoder input to the opposite COM port at the same baud rate.
6. Confirm the six icon PNG files are present in C:\MyData\PRAVE_ME_TimeKeeper\Icons\.
7. Start the field decoder first, then double-click the tester BAT file.
8. Allow at least one complete PRAVE cycle. With the default random timing, all six units normally appear within roughly 9-21 seconds.

8. Expected Google Earth Result

- One ME marker appears at the local RMC position.
- Six remote units appear around ME with IDs 7-101 through 7-106.
- 7-101 uses firetruck_rssi_unknown.png.
- 7-102 through 7-106 use firetruck_rssi_1.png through firetruck_rssi_5.png respectively.
- The exact numeric RSSI remains available in each placemark description and decoder log.
- The icons remain in place while their last-received times update as new PRAVE sentences arrive.

9. Check-Off Verification

[ ] Python self-test reports seven valid NMEA checksums.

[ ] Tester opens the configured output COM port at 19200 baud.

[ ] Decoder receives one GPRMC every second without losing PRAVE sentences.

[ ] ME appears at the expected center position.

[ ] All six PRAVE IDs appear after one shuffled cycle.

[ ] Unknown RSSI selects the unknown truck.

[ ] Each numeric RSSI selects the correct 1-5 bar truck.

[ ] PRAVE coordinates and IDs decode correctly.

[ ] RMC date/time is accepted as valid UTC.

[ ] Unplug/reconnect recovery works on the chosen serial hardware.

[ ] Activity logs preserve RMC and PRAVE transmission evidence.

10. Built-In Self-Test

The Python file can verify sentence generation without opening a COM port:

```
python PRAVE_RMC_Field_Decoder_Tester_v1_0_Gold.py --self-test
```

A passing result prints one valid RMC sentence and six valid PRAVE sentences, then reports seven successful checksums.

11. Troubleshooting

Symptom	Action
Could not open COM port	Confirm the COM number in Device Manager and make sure no other program has the tester port open.
No decoder traffic	Confirm both ends use the same baud rate and the serial path is null-modem/crossed correctly.
Bad checksum	Run --self-test. Do not manually edit generated sentence bodies below HERE BE DRAGONS.
Only some icons appear	Wait for a full shuffled cycle and verify the corresponding PNG filenames exist exactly.
Unknown unit does not use unknown icon	Confirm the decoder treats a blank PRAVE RSSI field as unknown rather than rejecting the whole valid sentence.
Windows time changes unexpectedly	The tester sends the tester computer's current UTC. Verify that computer's date/time before enabling decoder clock correction.

12. Lineage and Gold Rule

This tester is the direct bench successor to the earlier PRAVE Test Generator. It retains valid PRAVE checksum generation and the proven district/individual ID placement, then adds the 1 Hz GPRMC stream, the official RSSI field at split index 12, six simultaneous RSSI states, randomized PRAVE arrival timing, logging, auto-reconnect, and a no-hardware checksum self-test.

Gold Rule: Change the settings at the top. Leave the dragons alone.